

6.1 Group 2 (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.1 Group 2 (A Level Only)
Difficulty	Medium

Time allowed: 50

Score: /40

Percentage: /100

Question 1a

This question is about Group 2 nitrates.

Strontium nitrate is a Group 2 nitrate used to produce a colour in fireworks.

i)

Give the formula of strontium nitrate.

[1]

ii)

Give the colour that strontium nitrate will give in fireworks.

[1]

[2 marks]

Question 1b

i)

Write an equation for the reaction that occurs when strontium nitrate is heated.

You should include state symbols.

[2]

ii)

Give **one** observation of this reaction.

[1]

[1 mark]

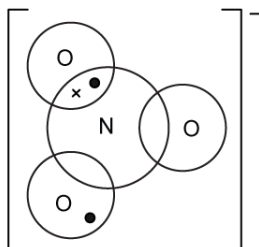
Question 1c

The nitrate ion, NO_3^- , contains a dative covalent bond.

Complete the following 'dot-and-cross' diagram of the bonding in the nitrate ion.

Use the following code for your electrons:

- electrons from oxygen
- x electrons from nitrogen
- $\square$ added electron(s) responsible for the overall negative charge



[3 marks]

Question 1d

Describe and explain the trend in thermal stabilities of the nitrates of the Group 2 elements.

[1 mark]

Question 2a

Magnesium is a Group 2 metal.

Write an equation, including state symbols for the reaction between:

i)
magnesium and steam

[1]

ii)
magnesium and sulfuric acid

[1]

[2 marks]**Question 2b**

Magnesium sulfate is soluble in water.

Describe and explain how the solubilities of the sulfates of the Group II elements vary down the group.

[4 marks]

Question 2c

The following table lists some enthalpy changes for magnesium and strontium compounds

Enthalpy change	Value for magnesium / kJ mol^{-1}	Value for strontium / kJ mol^{-1}
lattice enthalpy of $\text{M}(\text{OH})_2$	-2993	-2467
enthalpy change of hydration of $\text{M}^{2+}(\text{g})$	-1890	-1414
enthalpy change of hydration of $\text{OH}^-(\text{g})$	-550	-550

Use the above data to calculate values of $\Delta H^\circ_{\text{sol}}$ for $\text{Mg}(\text{OH})_2$ and for $\text{Sr}(\text{OH})_2$.

$\text{Mg}(\text{OH})_2$ kJ mol^{-1}

$\text{Sr}(\text{OH})_2$ kJ mol^{-1}

[2 marks]

Question 2d

Use your results from part (c) to suggest whether $\text{Sr}(\text{OH})_2$ is more or less soluble in water than $\text{Mg}(\text{OH})_2$, assuming all other factors are the same.

[2 marks]

Question 3a

This question is about Group 2 carbonates.

Group 2 carbonates can react with acids.

i)

Compare the reaction of calcium carbonate with hydrochloric acid and sulfuric acid.

[2]

ii)

Write an equation for the reaction between calcium carbonate and sulfuric acid, including state symbols.

[1]

[3 marks]

Question 3b

Describe and explain the trend observed in the thermal stability of the carbonates of the Group II elements.

[3 marks]

Question 3c

The ionic radii of three ions are shown in Table 4.1.

Suggest how the thermal stabilities of zinc carbonate and lead carbonate might compare to that of calcium carbonate.

Table 4.1

Ion	Ionic radii (nm)
Ca^{2+}	0.099
Zn^{2+}	0.074
Pb^{2+}	0.120

[2 marks]**Question 3d**

Zinc is found in d block of the Periodic table but is not a transition element.

i)

Explain why zinc is not classed as a transition element.

[1]

ii)

Describe three characteristic chemical properties of transition elements that are not shown by Group 2 elements.

[3]**[4 marks]**

Question 4a

This question is about Group 2 hydroxides.

Samples of magnesium, calcium, strontium and barium are reacted with water to form their hydroxides. The resulting solutions are then filtered to collect the precipitates.

Explain the trend in the expected mass for the precipitates.

[2 marks]

Question 4b

Table 4.1 shows the solubility data for the Group 2 metal hydroxides.

Table 4.1

Group 2 metal hydroxide	Solubility / g dm ⁻³ at 20 °C
Magnesium hydroxide	0.140
Calcium hydroxide	1.730
Strontium hydroxide	17.70
Barium hydroxide	38.90

A student determined that a 50 cm³ solution of an unknown Group 2 metal hydroxide contained 802 mg of the metal hydroxide.

Identify the metal hydroxide in the unknown sample.

[2 marks]

Question 4c

Write an equation for the reaction between magnesium hydroxide and sulfuric acid.

You should include state symbols.

[2 marks]

Question 4d

The solubility of Group 2 hydroxides increases going down the group because the enthalpy of solution, $\Delta H^\theta_{\text{sol}}$, gets more exothermic.

i)

Define the term enthalpy of solution, $\Delta H^\theta_{\text{sol}}$.

[2]

ii)

Explain why going down Group 2 the enthalpy of solution, $\Delta H^\theta_{\text{sol}}$ gets more exothermic.

[3]

[5 marks]

Model Answers: Medium

1a

a)

i) The formula of strontium nitrate is:

- $\text{Sr}(\text{NO}_3)_2$; [1 mark]

ii) The colour it will produce in a firework is:

- Red / scarlet; [1 mark]

[Total: 2 marks]

- You are expected to know the formula for Group 2 hydroxides, sulfates, nitrates and carbonates
 - Elements in Group 2 forms ions with a 2+ charge
 - Nitrate ions have a 1- charge
 - For the compound to have no charge overall, two nitrate ions are needed for every one strontium ion
- The strontium nitrate will produce a red colour due to the strontium ion
- You will not score the mark for giving the colour as brick red, as this colour is produced by Ca^{2+}

1b

b)

i) An equation for the reaction that occurs when strontium nitrate is heated is:



- Correct formulae; [1 mark]
- Correctly balanced; [1 mark]

ii) One observation of this reaction is:

Any **one** from:

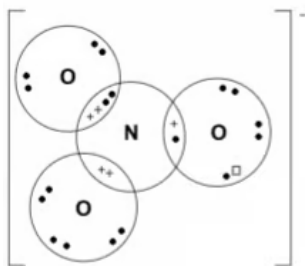
- Brown gas; [1 mark]
- White precipitate; [1 mark]

[Total: 3 marks]

- Thermal decomposition of nitrates produces a metal oxide, nitrogen dioxide and oxygen gas
- The nitrogen dioxide is observed as a brown gas
- The strontium oxide as a white precipitate

1c

c) The completed dot and cross diagram is:



- Dative covalent bond to an oxygen atom using two N electrons; [1 mark]
- 8 electrons around N in 1 double and 2 single bonds; [1 mark]
- A total of 24 electrons, including one $\square$; [1 mark]

[Total: 3 marks]

- Don't be put off completing a dot and cross diagram for an ion
- The single negative charge means an electron has been gained and you must use $\square$ to represent it or you will not score mark 3
- Each oxygen has six electrons in the outer shell
- Nitrogen has five electrons in the outer shell
- Arrange the electrons so that each atom has eight
- A dative covalent bond is one in which both electrons are donated by one atom, this must be represented using two of the symbol 'x'

1d

d) The trend in thermal stabilities of the nitrates of the Group 2 elements is:

- Thermal stability increases / Nitrates become more stable going down Group 2; [1 mark]
- (Because) the size / radius of the cation increases
OR
Charge density of the ion decreases; [1 mark]
- So polarisation / distortion of anion / nitrate ion decreases; [1 mark]

[Total: 3 marks]

- You can deduce that 1 mark must be for the description and 2 marks for the explanation
- At A level you must be able to explain the trends in the thermal stabilities of Group 2 carbonates and nitrates down the group
- The explanation for both is the same:
 - As the size of the Group 2 ion increases, its polarising power decreases
 - A decrease in the polarisation of the nitrate / carbonate ions means it is harder to weaken the N-O / C-O bond so the compound does not decompose

2a

a) An equation for the reaction between:

i) Magnesium and steam is:

- $\text{Mg (s)} + \text{H}_2\text{O (l)} \rightarrow \text{MgO (s)} + \text{H}_2\text{(g)}$; [1 mark]

ii) Magnesium and sulfuric acid is:

- $\text{Mg (s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)}$; [1 mark]

[Total: 2 marks]

- You are expected to be able to write equations for the reactions of Group 2 elements with oxygen, water, dilute HCl and dilute H_2SO_4
- **Remember:** Metal + acid $\rightarrow$ salt + hydrogen

2b

b) Going down Group 2:

- Solubility decreases / the sulfates get less soluble; [1 mark]
- (Because) lattice energy and hydration energies both decrease
OR
Lattice energy and hydration enthalpies get less negative / exothermic; [1 mark]
- The hydration enthalpy decreases more than the lattice energy
OR
The change in hydration enthalpy outweighs the change in lattice energy; [1 mark]
- So $\Delta H^\theta_{\text{sol}}$ gets more endothermic; [1 mark]

[Total: 4 marks]

- **Remember:** Although enthalpy changes of solution can be positive (endothermic) or negative (exothermic), a compound is likely to be soluble only if $\Delta H^\theta_{\text{sol}}$ is negative or has a small positive value
- Substances with large positive values for $\Delta H^\theta_{\text{sol}}$ are relatively insoluble so the value for $\Delta H^\theta_{\text{sol}}$ must increase going down Group 2

2c

c) The $\Delta H^\theta_{\text{sol}}$ for $\text{Mg}(\text{OH})_2$ and for $\text{Sr}(\text{OH})_2$ are:

- $\text{Mg}(\text{OH})_2$: $+3 \text{ kJ mol}^{-1}$; [1 mark]
- $\text{Sr}(\text{OH})_2$: -47 kJ mol^{-1} ; [1 mark]

[Total: 2 marks]

- To calculate the enthalpy of solution use: $\Delta H^\theta_{\text{sol}} = \Delta H^\theta_{\text{hyd}} - \Delta H^\theta_{\text{latt}}$
- For $\text{Mg}(\text{OH})_2$:
 - $\Delta H^\theta_{\text{sol}} = (-1890 + (-550 \times 2)) - (-2993)$
 - $\Delta H^\theta_{\text{sol}} = +3 \text{ kJ mol}^{-1}$
- For $\text{Sr}(\text{OH})_2$
 - $\Delta H^\theta_{\text{sol}} = (-1414 + (-550 \times 2)) - (-2467)$
 - $\Delta H^\theta_{\text{sol}} = -47 \text{ kJ mol}^{-1}$
- In both cases, you must multiply the enthalpy of hydration for the hydroxide ion by 2 as there are two ions present in both compounds

2d

d) $\text{Sr}(\text{OH})_2$ should be

- More soluble

ANDBecause $\Delta H^\theta_{\text{sol}}$ is more exothermic / negative; [1 mark]**[Total: 1 mark]**

- As discussed in part (b) the more exothermic the value of $\Delta H^\theta_{\text{sol}}$, the more soluble the substance is
- The trend in solubility of the Group 2 hydroxides is the opposite to that of the sulfates
- Group 2 hydroxide solubility increases going down the group because there is a larger decrease in $\Delta H^\theta_{\text{latt}}$ compared to $\Delta H^\theta_{\text{hyd}}$ (for sulfates there is a larger decrease in $\Delta H^\theta_{\text{hyd}}$ compared to $\Delta H^\theta_{\text{latt}}$)

3a

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2d

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3a

a)

i) Compare the reaction of magnesium carbonate with hydrochloric acid sulfuric acid:

- (Calcium carbonate reacts with both acids to) form a salt, carbon dioxide and water; [1 mark]
 - (But) calcium carbonate forms a soluble salt with hydrochloric acid
- AND**
- Calcium carbonate forms an insoluble salt (on top of the carbonate); [1 mark]

ii) An equation for the reaction between calcium carbonate and sulfuric acid is:

- $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$; [1 mark]

[Total: 3 marks]

- The command word 'compare' means to give similarities and differences

- If the question is 2 marks it is logical to deduce that you need one of each
- When a metal carbonate reacts with an acid:
 - Metal carbonate + acid $\rightarrow$ salt + water + carbon dioxide
- You need to link your knowledge of the solubility of Group 2 sulfates to this question:
 - Group 2 chlorides are all soluble
 - $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
 - Group 2 sulfates decrease in solubility going down Group 2
 - Magnesium sulfate is soluble in water but the elements further down Group 2 are not
 - Calcium actually forms an insoluble sulfate layer on top of the solid carbonates
 - You would have been awarded the mark had you said this for marking point 2
 - Make sure that the state symbol for calcium sulfate is (s) not (aq)

3b

b) Describe and explain the trend observed in the thermal stability of the carbonates of the Group II elements:

- Thermal stability increases down the group; [1 mark]
 - (Because) the cation radius increases
- OR**
- (Because) the charge density decreases; [1 mark]
- There is less polarisation / distortion of the carbonate ion / anion / CO_3^{2-} ; [1 mark]

[Total: 3 marks]

- There are very few ways you can be asked about the thermal stability of the carbonates and nitrates so it is worth learning this explanation
- The greater the polarisation of the carbonate ion, the easier it is to weak the carbon-oxygen bond and decompose the compound
- A carbonate ion is polarised more by cations with a smaller ionic radius
- Going down Group 2, the ionic radius of the cations increases so those ions at the bottom of the group cannot polarise the carbonate ion as much as those at the top resulting in greater thermal stability

3c

c) Suggest how the thermal stabilities of zinc carbonate and lead carbonate might compare to that of calcium carbonate:

- Zinc carbonate is less thermally stable / has lower thermal stability

AND

Because the ionic radius (of the cation) is smaller; [1 mark]

- Lead carbonate is more thermally stable / has lower thermal stability

AND

Because the ionic radius (of the cation) is greater; [1 mark]

[Total: 2 marks]

- **Remember:** The smaller the cation, the better it is at polarising the carbonate ion resulting in easier breakage of the C-O bond, and the compound decomposing
- You do not need to be able to remember the trend in thermal stabilities of elements other than those of Group 2 nitrates and carbonates but you should be able apply your understanding of why thermal stabilities change to a different context, such as this one

3d

d)

i) Zinc is not classed as a transition element because:

- It forms an ion with complete (3d) subshell; [1 mark]

ii) **Three** characteristic chemical properties of transition elements that are not shown by Group 2 elements are:

Any **three** from:

- Form coloured ions / compounds; [1 mark]
- Variable oxidation states; [1 mark]
- Forms complexes; [1 mark]
- Act / behave / used as catalysts; [1 mark]

[Total: 4 marks]

- Transition elements are elements with an incomplete d-subshell that can form at least one stable ion with an incomplete d-subshell
- Zinc has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$
- It forms only one ion, Zn^{2+} which has the electronic configuration, $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$
 - The 3d subshell is complete

4a

a) The trend in the expected mass for the precipitates is:

Alternative 1:

- Magnesium hydroxide will have the highest / greatest mass (of precipitate)
AND
The mass (of precipitate) will decrease as you move down the group; [1 mark]

Alternative 2:

- Barium hydroxide will have the lowest / smallest mass (of precipitate)
AND
The mass (of precipitate) will increase as you move up the group; [1 mark]
- (Because) solubility of Group 2 hydroxides increases going down Group 2; [1 mark]

[Total: 2 marks]

- This question is about the solubility of the Group 2 metal hydroxides in water
- **Careful:** the talk about precipitates in the question is to distract you
 - Magnesium hydroxide is sparingly soluble in water so there will be a large amount of precipitate
 - **Solubility of the hydroxides increases** as you move **down the group**
 - Barium hydroxide is very soluble in water so there will be a small amount of precipitate, if any

4b

b) To identify the metal hydroxide in the unknown sample:

- $(802 / 1000) = 0.802 \text{ g (in } 50 \text{ cm}^3)$

AND

$$(0.802 \times 20) = 16.04 \text{ g dm}^{-3} \text{ (in } 1000 \text{ cm}^3); [1 \text{ mark}]$$

- (The unknown metal hydroxide is) strontium hydroxide; [1 mark]

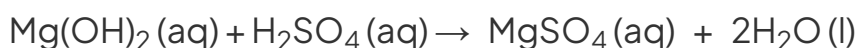
[Total: 2 marks]

- **Careful:** The information in the table wants you to work in g dm^{-3} but you have information in mg and cm^3
 - $1 \text{ g} = 1000 \text{ mg}$
 - $1 \text{ dm}^3 = 1000 \text{ cm}^3$
- To convert the mass from mg to g , divide by 1000
- To scale the mass according to the volume $\times 20$
- Use your answer to identify the correct metal hydroxide from the table by choosing the closest value
- There will always be differences between experimental values and data book tables
 - In this question, the main reasons for the difference could be due to the temperature of the water and impurities in the unknown sample

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- There will always be differences between experimental values and data book tables
 - In this question, the main reasons for the difference could be due to the temperature of the water and impurities in the unknown sample
- You cannot achieve the mark for identifying the metal unless you have shown your calculation detailing how you arrived at that answer

4c

c) An equation for the reaction between magnesium hydroxide and sulfuric acid is:



- Balanced equation; [1 mark]
- Correct species; [1 mark]

[Total: 2 marks]

- Make sure you have the correct state symbol for magnesium sulfate, it is at the top of Group 2
 - The solubility of sulfates decreases going down the group

4d

d)

i) The term enthalpy of solution means:

- The enthalpy change when 1 mole of ionic solid; [1 mark]
- dissolves to form a dilute solution under standard conditions; [1 mark]

ii) Going down Group 2 the enthalpy of solution, $\Delta H^\theta_{\text{sol}}$ gets more exothermic because:

- The enthalpy change of hydration decreases (down the group)

OR

The enthalpy change of hydration gets more endothermic (down the group); [1 mark]

- The lattice energy decreases (down the group)

OR

The lattice energy gets more endothermic (down the group); [1 mark]

- The decrease in lattice energy outweighs the decrease in enthalpy change of hydration; [1 mark]

[Total: 5 marks]

- For small negative ions like hydroxide, the lattice enthalpy falls faster than the hydration enthalpy of the positive ions
- That means that the enthalpy of solution will become less positive (or more negative)
- This is the opposite for larger negative ions like sulfate or carbonate
- For these, the hydration enthalpy of the positive ions falls faster than the lattice enthalpy
- That means the enthalpy of solution will become more positive (or less negative)